

CoGD-Net: Conditional Graph Diffusion for Long-Tail Object Segmentation with Privileged Priors

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Abstract

Precise segmentation of diminutive (≤ 5 mm) and low-contrast lesions remains a significant challenge in medical imaging, as standard models often underperform on the long tail of small targets. We propose **CoGD-Net**, a framework that formulates segmentation refinement as a *Conditional Graph Diffusion* (CoGD) process. By modeling latent feature tokens as nodes in a dynamic graph, we implement a learnable reaction-diffusion mechanism that is explicitly modulated by task-specific priors. To leverage rich training data without increasing deployment complexity, we introduce *Privileged Information Distillation*. This strategy aligns the diffusion dynamics using paired multimodal data during training, while maintaining robust single-modality inference through stochastic dropout. In evaluations across four benchmarks, CoGD-Net achieves state-of-the-art results for diminutive polyp segmentation. Our analysis suggests that embedding structured priors into a learnable diffusion mechanism is highly effective for improving robustness in long-tailed medical vision tasks.

1. Introduction

Accurate detection and segmentation of diminutive (≤ 5 mm) and low-contrast objects is a critical yet unresolved problem in computer vision, particularly in safety-critical domains such as computer-aided colonoscopy. While state-of-the-art models show high performance on common, well-defined targets, they frequently struggle with the *long tail* of small or visually ambiguous instances (Cui et al., 2019). In clinical practice, missing a diminutive precancerous polyp during screening can lead to delayed diagnosis and increased cancer risk (Jha et al., 2020; Silva et al., 2014).

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Current segmentation frameworks—ranging from classical U-Net variants (Ronneberger et al., 2015; Fan et al., 2020) to Vision Transformers (Xie et al., 2021; Duc et al., 2022)—predominantly optimize for average overlap metrics like Dice or IoU. These metrics are inherently biased toward larger objects, often resulting in low recall for smaller targets. Furthermore, most systems rely on a single input modality (e.g., white-light imaging), failing to exploit potentially rich multimodal cues (e.g., narrow-band imaging) or structured clinical metadata (e.g., predicted lesion size) during feature learning. While such auxiliary data may be unavailable at deployment, using it as *privileged information* during training could significantly enhance the model’s ability to resolve weak signals.

The recent success of foundation models like Segment Anything (SAM) (Kirillov et al., 2023) demonstrates powerful zero-shot capabilities. However, their performance on small, low-contrast targets remains suboptimal in fully automatic settings (Ma et al., 2023). These models typically employ generic prompts or implicit contextualization, often lacking a mechanism to systematically integrate task-specific physical or clinical priors to guide feature refinement for challenging long-tailed distributions.

To address these limitations, we introduce **CoGD-Net**, a framework centered on *Conditional Graph Diffusion*. We formulate the refinement of feature maps as a learnable, prior-modulated diffusion process in latent space. By representing feature maps as sparse graphs, CoGD-Net propagates information using edge weights that are *conditionally modulated* by lesion-aware priors. A key innovation is our *Privileged Information Distillation* strategy. During training, CoGD-Net leverages paired multimodal data (e.g., from PICCOLO (Sánchez-Peralta et al., 2020)) to align graph representations. Crucially, a stochastic modality dropout mechanism ensures the model remains robust when only standard single-modality inputs are available at test time.

In this work, we focus on explicitly embedding geometric and clinical priors into the feature refinement process. Our primary contributions are:

- We propose **CoGD-Net**, which conceptualizes feature refinement as a parameterized reaction-diffusion sys-

tem on graphs.

- We introduce a *Privileged Information Distillation* strategy that combines cross-modal prototype alignment with stochastic dropout to facilitate robust single-modality inference.
- We present a size-stratified evaluation across four benchmarks (PICCOLO, Kvasir-SEG, CVC-ClinicDB, ETIS-Larib), showing that CoGD-Net significantly improves recall for diminutive lesions (<5 mm) and enhances cross-dataset generalization.

These results advocate for actively sculpting representation space geometry using clinical priors, specifically targeting the long-tail regime of hard-to-detect objects.

2. Related Work

Long-tail recognition and diminutive object segmentation. The challenge of long-tailed data distributions, where performance is often bottlenecked by rare classes or diminutive instances, is widely recognized in computer vision (Cui et al., 2019). For dense prediction, this manifests as poor recall for small objects. Approaches include re-sampling, re-weighting losses, or group-distributionally robust optimization (Sagawa et al., 2020). Polyp segmentation is a critical instance of this problem (Fan et al., 2020; Huang et al., 2021), with methods ranging from CNNs (Ronneberger et al., 2015) to Transformers (Xie et al., 2021; Duc et al., 2022) using architectural tweaks like FPNs (Lin et al., 2017). However, these rarely exploit explicit structured priors when visual evidence is weakest, precisely where diminutive lesions fall. Our work offers a complementary, mechanism-driven solution.

Promptable foundation models and multimodal learning. Foundation models like SAM (Kirillov et al., 2023) and its variants (Ravi et al., 2024; Camarillo et al., 2025; Ma et al., 2023) excel in interactive segmentation tasks, leveraging generic prompts. Yet, their fully automatic performance on challenging small, low-contrast objects can be limited, as they lack task-specific inductive biases. Multimodal learning, through frameworks like CLIP (Radford et al., 2021) and its medical adaptations (Zhang et al., 2023; Lin et al., 2023), offers robust cross-modal representations. However, these typically focus on classification or high-level alignment. We advance multimodal learning by using multimodal cues as “privileged information” during training to explicitly condition latent feature dynamics for pixel-level tasks, with robust single-modality deployment.

Graph neural networks and neural operators for feature propagation. Graph neural networks (GNNs) (Kipf & Welling, 2017; Veličković et al., 2018) and continuous-time

neural ODEs (Chen et al., 2018) model feature propagation on irregular domains. Diffusion models (Ho et al., 2020) and neural operators (Li et al., 2021) approximate PDE-like dynamics for iterative refinement. Our work draws inspiration from these, but significantly differs by introducing a *conditional* graph diffusion that is explicitly modulated by task-specific priors, rather than generic smoothing.

Geometric and Physical Priors in Medical Vision. Unlike natural images, medical data often adheres to underlying physical or geometric constraints. Recent trends involve integrating PDE-based evolution or graph-based topology constraints to regularize segmentation. Our work aligns with this *Geometric Deep Learning* perspective, models feature evolution as a heat diffusion process on a manifold defined by image semantics, rather than treating the image merely as a grid of pixels.

Learning with Privileged Information. Our approach embraces the Learning Using Privileged Information (LUPI) paradigm, which leverages additional data available only during training to improve test-time performance. In medical imaging, multimodal data (like NBI) often serves as this privileged information. CoGD-Net adopts this via our distillation strategy, ensuring the model benefits from auxiliary signals during key training phases without creating a deployment dependency on them.

3. Theoretical Motivation: Latent Reaction-Diffusion on Graphs

The problem of segmenting diminutive, low-contrast objects often stems from their weak visual signals easily being attenuated or corrupted during feature extraction. We postulate that this can be addressed by guiding feature propagation in latent space using a parameterized diffusion process.

Consider a feature map flattened into tokens $X \in \mathbb{R}^{N \times C}$. We view these tokens as nodes on a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Our objective is to refine these features $\mathbf{h} \in \mathbb{R}^{N \times C'}$ through a process inspired by reaction-diffusion systems. In continuous time, the evolution of features on a graph can be analogized to a system of coupled PDEs:

$$\frac{\partial \mathbf{h}(t)}{\partial t} = \mathcal{D}(c, \mathbf{h}(t))\mathbf{h}(t) + \mathcal{S}(c, \mathbf{h}(t)) \quad (1)$$

where $\mathcal{D}$ represents a diffusion operator (analogous to a graph Laplacian), and $\mathcal{S}$ represents a source/reaction term. Both $\mathcal{D}$ and $\mathcal{S}$ are *conditioned* on a prior vector c , allowing the dynamics to adapt to specific object characteristics (e.g., predicted size, modality).

CoGD-Net provides a learnable, discrete-time approximation to such a system. The conditioned diffusion operator $\mathcal{D}(c, \mathbf{h})$ is realized through our modulated graph Laplacian

$\tilde{L}(c, \mathbf{h})$, while the absence of an explicit source term (other than the initial feature projection) implicitly means the diffusion is controlled to aggregate existing weak signals. This allows CoGD-Net to selectively amplify subtle lesion evidence in regions where priors suggest a diminutive object, preventing over-smoothing in areas with strong signals, and actively shaping the feature geometry to enhance long-tail detectability.

4. Method

4.1. Problem Setup

We address dense prediction from endoscopic images, potentially with multi-modal and clinical priors. For case i : x_i^{wli} is a white-light (WLI) frame, x_i^{nbi} an optional narrow-band imaging (NBI) frame, and a_i case-level attributes (e.g., size, pathology). The ground-truth is y_i . Our model f_θ predicts $\hat{y}_i = f_\theta(x_i^{\text{wli}}, x_i^{\text{nbi}}, a_i)$. We focus on *diminutive* or long-tail lesions (e.g., ≤ 5 mm). Our core strategy involves training f_θ using multimodal inputs and auxiliary priors (privileged information), while ensuring robust operation using only WLI at inference (train-multimodal / test-unimodal).

4.2. Overview of CoGD-Net

CoGD-Net is a plug-in architecture augmenting a strong segmentation backbone with a Conditional Graph Diffusion (CoGD) block (Fig. 1). The backbone, typically a Transformer, extracts multi-scale features. At a selected bottleneck scale, feature maps are flattened into a sparse token graph. CoGD then performs a few steps of learnable diffusion on this graph. Both diffusion strength and edge weights are dynamically modulated by a low-dimensional condition vector c encoding modality and clinical priors. This adaptive diffusion propagates information more aggressively where diminutive or low-contrast lesions are expected, effectively amplifying weak signals. Diffused features are then reshaped and fed to the segmentation head.

During training, we leverage paired WLI-NBI views and metadata for cross-modal alignment and randomly drop auxiliary modalities to ensure unimodal WLI-only inference without architecture changes.

4.3. Backbone and Multi-scale Features

We use a transformer-based encoder (e.g., SegFormer (Xie et al., 2021)) to produce multi-scale features $\{F^s\}_{s=1}^S$, where $F^s \in \mathbb{R}^{H_s \times W_s \times C_s}$. CoGD is applied to a single intermediate scale s^* .

4.4. Sparse Graph Construction

Tokens at scale s^* are flattened to $X \in \mathbb{R}^{N \times C}$, $N = HW$. A sparse pixel graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with: (i) **Spa-**

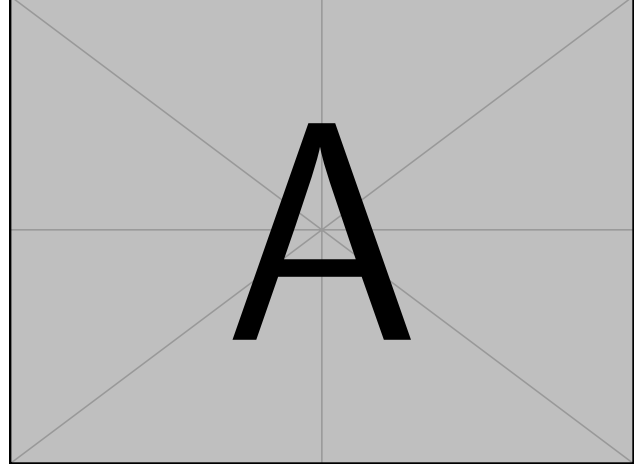


Figure 1. Overview of CoGD-Net. A strong polyp segmentation backbone is augmented with conditional graph diffusion that injects multi-modal priors to boost recall on small lesions.

tial edges to 4- or 8-neighbors, preserving locality; (ii) **Feature kNN edges** ($k = 8$) using cosine similarity, connecting semantically related tokens. Let A be the weighted adjacency (feature edges weighted by $\exp(\text{sim}(\mathbf{x}_i, \mathbf{x}_j))$, spatial edges set to 1, then row-normalized), D its degree matrix. The normalized graph Laplacian is defined as

$$L = I - D^{-1/2} A D^{-1/2}, \quad (2)$$

following (Kipf & Welling, 2017). The condition vector $c \in \mathbb{R}^{d_c}$ (encoding modality, size bin, pathology priors) is mapped by a small MLP $\phi(c)$ to modulate diffusion.

4.5. Conditional Graph Diffusion (CoGD)

CoGD performs T explicit diffusion steps, conditioned on c :

$$\begin{aligned} H_0 &= \text{Proj}(X), \\ H_{t+1} &= H_t - \alpha_t(c) \tilde{L}(c, H_t) H_t, \end{aligned} \quad (3)$$

where $H_t \in \mathbb{R}^{N \times C'}$ and Proj is a linear projection. The step size $\alpha_t(c) \in (0, \alpha_{\max}]$ is produced by a two-layer MLP with a sigmoid output, allowing conditional control of diffusion strength. This process realizes a discrete approximation of the latent reaction-diffusion system (Sec. 3). Crucially, this discrete diffusion step is fully differentiable. The modulated Laplacian $\tilde{L}(c, H_t)$ serves as a learnable transition matrix where gradients flow not only through the node features H_t but also through the condition vector c and the gating network ϕ . This allows the model to learn the optimal “conductivity” of the feature graph end-to-end, actively amplifying signals in regions identified as high-priority by the priors.

Conditioned Edge Gating. To enable anisotropic, signal-aware diffusion, edges are reweighted at each step using a FiLM-style gate (Perez et al., 2018):

$$\tilde{A}_{ij} = A_{ij} \cdot g_{\phi}(c, h_{i,t}, h_{j,t}), \quad g_{\phi}(\cdot) = \sigma(w^{\top} [h_{i,t}; h_{j,t}; \phi(c)]), \quad (4)$$

from which $\tilde{L}$ is derived as in Eq. (2). This allows c to dynamically shape the effective graph topology and diffusivity, e.g., enabling stronger information aggregation along semantically plausible edges when small, low-contrast polyps (weak evidence) are anticipated, thereby enhancing their signal-to-noise ratio. With $|E| = \mathcal{O}(Nk)$, CoGD’s complexity is $\mathcal{O}(T|E|C')$, remaining lightweight for $T \leq 5$.

4.6. Segmentation Head

The final diffused features are reshaped back into a feature map and fused with other scales via a standard decoder: $\hat{y} = \text{Head}(\text{Reshape}(H_T), \{F^s\}_{s \neq s^*})$, producing $\hat{y} \in [0, 1]^{H_0 \times W_0}$.

4.7. Privileged Information Distillation (Train-MM/Test-UM)

PICCOLO (Sánchez-Peralta et al., 2020) provides paired WLI and NBI frames with metadata, serving as *privileged information* during training. At inference, only WLI is guaranteed.

Prototype-level Contrastive Alignment. For an image x^m from modality $m \in \{\text{wli}, \text{nbi}\}$, we compute a lesion-aware prototype $p^m = \text{Pool}(H_T(x^m)) \in \mathbb{R}^{C'}$ (masked global pooling over lesion region). We use a CLIP-style contrastive loss (Radford et al., 2021) to align prototypes from the same polyp: $\mathcal{L}_{\text{align}} = -\log \frac{\exp(\text{sim}(p^{\text{wli}}, p^{\text{nbi}})/\tau)}{\sum_k \exp(\text{sim}(p^{\text{wli}}, p^k)/\tau)}$, where k iterates over batch prototypes. This distills cross-modal consistency into the latent feature space.

Stochastic Privileged Information Dropout. To guarantee the model functions reliably with only standard imaging, we randomly mask out the NBI views and metadata during training. We use a Bernoulli mask ($p = 0.3$) on these auxiliary pathways. This forces the network to infer the missing contextual priors from the white-light input alone, effectively preventing over-reliance on the “privileged” signals.

4.8. Objectives: Segmentation, Tail Robustness and Alignment

We optimize $\mathcal{L} = \mathcal{L}_{\text{seg}} + \beta \mathcal{L}_{\text{align}} + \gamma \mathcal{L}_{\text{tail}}$. $\mathcal{L}_{\text{seg}} = \lambda_1 \text{BCE}(\hat{y}, y) + \lambda_2 (1 - \text{Dice}(\hat{y}, y))$. For $\mathcal{L}_{\text{tail}}$, we define size bins and use size-bin reweighting or GroupDRO (Sagawa et al., 2020) to focus on diminutive lesions.

Algorithm 1 CoGD-Net Training (Train-MM / Test-UM)

```

1: Input: image(s)  $\{x^m\}$ , mask  $y$ , condition  $c$ 
2: Encode features  $\{F^s\} \leftarrow \text{Backbone}(x)$ 
3: Select scale  $s^*$  and form  $X \leftarrow \text{Flatten}(F^{s^*})$ 
4: Build sparse graph  $\mathcal{G}$ : spatial + kNN edges, Laplacian  $L$ 
5:  $H_0 \leftarrow \text{Proj}(X)$ 
6: for  $t = 0$  to  $T - 1$  do
7:    $\tilde{A} \leftarrow A \odot g_{\phi}(c, H_t)$ ,  $\tilde{L} \leftarrow \text{Norm}(\tilde{A})$ 
8:    $H_{t+1} \leftarrow H_t - \alpha_t(c) \tilde{L} H_t$ 
9: end for
10:  $\hat{y} \leftarrow \text{Head}(\text{Reshape}(H_T), \{F^s\}_{s \neq s^*})$ 
11: Compute  $\mathcal{L}_{\text{seg}}$  and  $\mathcal{L}_{\text{tail}}$  (size-bin based)
12: If multimodal pair: compute prototypes and  $\mathcal{L}_{\text{align}}$ 
13: Update all parameters by  $\nabla(\mathcal{L}_{\text{seg}} + \beta \mathcal{L}_{\text{align}} + \gamma \mathcal{L}_{\text{tail}})$ 

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5. Experiments

5.1. Experimental Setup

Datasets. We evaluate CoGD-Net on Kvasir-SEG (Jha et al., 2020), CVC-ClinicDB (Bernal et al., 2017), ETIS-Larib (Silva et al., 2014), and PICCOLO (Sánchez-Peralta et al., 2020).

Baselines. We compare with: (1) Polyp-specific CNN/Transformer models (U-Net, PraNet, HarDNet-MSEG, ColonFormer); (2) Strong generic segmenters (SegFormer, Mask2Former); (3) Promptable foundation models (SAM, SAM 2, SAM 3, MedSAM).

5.2. Main Results

5.3. Small-Lesion Performance and ≤ 5 mm Analysis

PICCOLO: Millimeter-accurate lesion size. Leveraging PICCOLO’s (Sánchez-Peralta et al., 2020) mm-level metadata, Table 2 presents performance on ≤ 5 mm, 6–9 mm, and ≥ 10 mm bins. CoGD-Net achieves substantial gains in the ≤ 5 mm group, significantly improving $\text{Recall}_{\leq 5}$ and reducing $\text{FNR}_{\leq 5}$ compared to its SegFormer backbone, while maintaining competitive performance for larger lesions. This validates our core hypothesis: conditional diffusion along clinically informed structures effectively recovers micro-lesion signals.

5.4. Cross-dataset Generalization to ETIS

Unseen distribution shift. ETIS-Larib (Silva et al., 2014) is a challenging OOD benchmark. Models trained on Kvasir-SEG $\cup$ CVC-ClinicDB are evaluated zero-shot on ETIS. Table 3 shows CoGD-Net generalizes better than specialized CNNs and Transformer baselines, and is competitive with foundation models even with oracle prompts. These gains

Table 1. In-domain segmentation performance on Kvasir-SEG and CVC-ClinicDB. Numbers are reported as Dice / IoU (%). **Bold** indicates best.

Method	Kvasir-SEG		CVC-ClinicDB	
	Dice $\uparrow$	IoU $\uparrow$	Dice $\uparrow$	IoU $\uparrow$
U-Net	xx.x	xx.x	xx.x	xx.x
PraNet	xx.x	xx.x	xx.x	xx.x
HarDNet-MSEG	xx.x	xx.x	xx.x	xx.x
ColonFormer	xx.x	xx.x	xx.x	xx.x
SegFormer-B2	xx.x	xx.x	xx.x	xx.x
MedSAM	xx.x	xx.x	xx.x	xx.x
SAM 2 (oracle)	xx.x	xx.x	xx.x	xx.x
CoGD-Net (ours)	xx.x	xx.x	xx.x	xx.x

Table 2. Size-stratified performance on PICCOLO (Sánchez-Peralta et al., 2020). We report Dice / IoU (%) and lesion-level recall / FNR (%) for three mm-level bins.

Size bin	Dice $\uparrow$	IoU $\uparrow$	Recall $\uparrow$	FNR $\downarrow$
≤ 5 mm	xx.x	xx.x	xx.x	xx.x
6–9 mm	xx.x	xx.x	xx.x	xx.x
≥ 10 mm	xx.x	xx.x	xx.x	xx.x
CoGD-Net (ours)	xx.x	xx.x	xx.x	xx.x

Table 3. Cross-dataset generalization to ETIS-Larib (Silva et al., 2014). Models are trained on Kvasir-SEG $\cup$ CVC-ClinicDB and evaluated on ETIS without fine-tuning. We report Dice / IoU (%) on all lesions and on the *small* area-based bin.

Method	All lesions		Small lesions	
	Dice $\uparrow$	IoU $\uparrow$	Dice $\uparrow$	IoU $\uparrow$
PraNet	xx.x	xx.x	xx.x	xx.x
HarDNet-MSEG	xx.x	xx.x	xx.x	xx.x
SegFormer-B2	xx.x	xx.x	xx.x	xx.x
SAM 2 (oracle)	xx.x	xx.x	xx.x	xx.x
CoGD-Net (ours)	xx.x	xx.x	xx.x	xx.x

are significant for small lesions, indicating conditional diffusion counteracts distribution shifts affecting micro-lesions.

5.5. Ablation Studies

Effect of conditional diffusion. Table 4 ablates CoGD components. Unconditioned diffusion improves small-lesion recall but can cause over-smoothing. Conditional diffusion, however, recovers large-lesion performance and further boosts ≤ 5 mm recall, underscoring the necessity of prior-modulated dynamics.

5.6. Qualitative Analysis

Figure 2 showcases CoGD-Net’s ability to preserve tiny lesions and avoid spurious responses on challenging cases. Unlike traditional models or foundation models, CoGD-Net’s prior-modulated diffusion yields more coherent and

Table 4. Ablation on conditional graph diffusion and tail-aware learning. We report overall Dice (%) and small-lesion recall (%) on PICCOLO (≤ 5 mm).

Variant	Kvasir Dice	PICCOLO Dice	Recall $_{\leq 5}$
(a) Backbone only	xx.x	xx.x	xx.x
(b) + Diffusion	xx.x	xx.x	xx.x
(c) + CoGD	xx.x	xx.x	xx.x
(d) + Distillation	xx.x	xx.x	xx.x
Full Model	xx.x	xx.x	xx.x

Placeholder for Qualitative Examples

Figure 2. Qualitative comparison on diminutive and low-contrast lesions from the PICCOLO and ETIS datasets.

accurate predictions for very small, low-contrast targets.

6. Discussion

Our results confirm that conditional graph diffusion is most beneficial for diminutive targets, while maintaining strong performance on larger objects. The explicit, learned diffusion steps act as an adaptive, anisotropic feature aggregation on a token graph. Unlike uniform smoothing, this selectively amplifies weak but consistent cues within compact, semantically relevant neighborhoods, effectively boosting the signal-to-noise ratio of subtle features without crude size heuristics. The condition tokens c critically modulate step size and edge gating, enabling a learned control over the diffusivity tensor. Furthermore, multimodal alignment

strengthens feature representations by enforcing consistency across diverse observations of the same underlying object.

7. Conclusion

In this work, we introduced **CoGD-Net**, a novel architecture designed to address the challenges of segmenting diminutive and low-contrast lesions. By formulating feature refinement as a prior-modulated reaction-diffusion process on graphs, we move beyond conventional static feature aggregation. Our evaluations across multiple benchmarks demonstrate that CoGD-Net significantly improves the detection of micro-lesions while maintaining high overall segmentation accuracy. These results underscore the importance of integrating structured clinical and geometric priors to improve the robustness of medical imaging systems in long-tailed regimes.

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